



Mg–Sr–Nd isotopic constraints on the genesis of the giant Jinchuan Ni–Cu–(PGE) sulfide deposit, NW China

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ABSTRACT

Magmatic sulfide deposits are principal resources of nickel and platinum-group elements (PGE) in the world. These deposits form as the result of the concentration of immiscible sulfide-liquid droplets segregated from mafic or ultramafic magma. Sulfide saturation in the magmas is commonly linked to contamination by the continental crust. The Jinchuan Ni–Cu–(PGE) deposit is the largest single magmatic sulfide deposit in the world. It is hosted by a small mafic–ultramafic intrusion in the North China Craton. The Jinchuan mafic–ultramafic intrusion intruded Paleoproterozoic marbles, gneisses, and granitic migmatites. This paper reports the magnesium, strontium and neodymium isotopic compositions of the intrusion and the country rocks. Olivine, pyroxene and whole rocks of the lherzolites from the Jinchuan intrusion have $\delta^{26}\text{Mg}$ values ranging from -0.40‰ to -0.26‰ , with a weighted mean value of $-0.31 \pm 0.06\text{‰}$ (2SD), which is slightly lighter than expected in a mantle-derived magma. The marble country rocks have relatively light $\delta^{26}\text{Mg}$ values varying from -2.00‰ to -1.17‰ , with a weighted mean value of $-1.53 \pm 0.02\text{‰}$ (2SD). The variations are attributed to the interaction between basaltic magma and marble country rocks as the principal mechanism which generated the low Mg isotopic signatures in the Jinchuan deposit. The $(^{87}\text{Sr}/^{86}\text{Sr})_i$ and ϵ_{Nd} values of the lherzolites vary from 0.706615 to 0.714612 and from -8.29 to -6.13 , respectively. The marble country rocks have relatively high $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios (0.713662 to 0.740069) and low ϵ_{Nd} values (-25.43 to -12.41). The negative Nb anomalies in all samples from the intrusion indicate that the parental magma underwent crustal contamination. **Mg–Sr–Nd isotopic mixing calculations indicate that the Jinchuan intrusion underwent 10–20% bulk contamination involving the lower continental crust**, followed by $<10\%$ bulk contamination by marbles. Our petrological and geochemical investigation of the Jinchuan intrusion provides evidence that a combination of the two processes led to the formation of the ore deposit. The assimilation of lower crustal rocks reduced the sulfur solubility; the assimilation of marble country rocks made the magma more oxidized and hence decreased the sulfur solubility further. The results from this study show that Mg isotopes are useful in the study of carbonate assimilation by mafic magma, and marble assimilation by basaltic magma is an important mechanism to produce sulfide saturation in the genesis of Ni–Cu–PGE sulfide deposits.

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1. Introduction

Magmatic sulfide deposits, which account for approximately 60 percent of the world's Ni production and >90 percent of the world's Pt–Pd production, form as a result of the concentration of immiscible sulfide-liquid droplets segregated from mafic or ultramafic magma. The contamination of mantle-derived magmas by the continental crust is considered an important process contributing to magmatic differentiation and evolution of the magmatic

ore deposits (e.g. Naldrett, 2011; Ripley and Li, 2013). In most deposits, the assimilation of S-rich country rocks can promote sulfide saturation in a mafic magma; such a process was important for the genesis of the Noril'sk–Talnakh Ni–Cu–platinum-group elements (PGE) sulfide deposits (Naldrett, 2011). In addition, the assimilation of carbonate country rocks by the parent magma has been proposed to explain the genesis of certain petrological features of magmatic deposits worldwide (e.g. Wenzel et al., 2002; Holwell et al., 2007; Ganino et al., 2008, 2014; Tang et al., 2017), such as the Platreef Ni–Cu–PGE sulfide deposit of the Bushveld Complex, the Aguablanca Ni–Cu sulfide deposit and the Panzhihua Fe–Ti–V oxide deposit. The Jinchuan Ni–Cu–(PGE) sulfide deposit in northwestern China, which contains more than 500 mil-

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lion metric tons of sulfide ores with grades of 1.1 wt.% Ni and 0.7 wt.% Cu, is one of the three largest magmatic sulfide deposits in the world (Naldrett, 2011), and the largest single magmatic Ni–Cu sulfide ore deposit in the world (Li and Ripley, 2011; Zhang et al., 2013). The Jinchuan intrusion intruded the Proterozoic Baijiazui Formation which comprises marbles, gneisses, and granitic migmatites. The Jinchuan intrusion was part of the product of continental rift-related basaltic magmatism which took place during the early stage of Rodinia breakup (Tang et al., 2014). Re–Os–Hf–Sr–Nd and multiple S isotopes reveal that crustal contamination was important in the formation of the Jinchuan deposit (Yang et al., 2005, 2008; Song et al., 2006, 2012; Duan et al., 2016). Multiple S isotopes indicate that both Archean and post-Archean crustal sulfur were involved in the generation of the Jinchuan deposit (Duan et al., 2016). Marble xenoliths are common in the contact aureole along the base of the intrusion and dispersed through much of the intrusion (Lehmann et al., 2007; Song et al., 2012). On the basis of the geological observation and the variation of major elements, Lehmann et al. (2007) propose that magma interaction with carbonate-rich fluids may be linked to the formation of the Jinchuan Ni–Cu–(PGE) deposit. This hypothesis is difficult to evaluate using carbon and oxygen isotopes because CO_2 is transient in the magma due to extremely low solubility in mafic magma under upper crustal pressures. Additionally, carbonates tend to have low contents of radioactive U, Th, Lu and Sm, so commonly applied isotopic tracers of crustal assimilation such as $^3\text{He}/^4\text{He}$, Lu–Hf and Sm–Nd, are insensitive to carbonate assimilation. Recent studies have shown that magnesium (Mg) isotopes of igneous rocks are an effective tool in distinguishing carbonate contamination and the widespread disequilibrium fractionation during magmatic processes (Huang and Xiao, 2016; Ke et al., 2016; Chen et al., 2018). Magnesium is a major element in the mantle. The fractionation of Mg isotopes during partial melting and magma differentiation is limited in the mantle ($<0.07\%$ for $\delta^{26}\text{Mg}$, Teng et al., 2010). Compared with the uniform $\delta^{26}\text{Mg}$ ratios in the upper mantle ($\delta^{26}\text{Mg} = -0.25 \pm 0.07\%$, 2SD; Teng et al., 2010), clastic sedimentary rocks may have slightly heavy Mg isotopic compositions varying from -1.64% to $+0.92\%$ (Li et al., 2010), whereas carbonates have highly variable light Mg isotopic compositions ranging from -5.60% to -0.66% (Young and Galy, 2004; Beinlich et al., 2014). Therefore, Mg isotope compositions can be sensitive tracers of contamination of mantle-derived magma by crustal material (Brewer et al., 2018).

In this paper, we present the Mg isotopic compositions of mineral separates and whole rock samples from the giant Jinchuan Ni–Cu–(PGE) sulfide deposit and the marble country rocks, along with major, trace element and Sr–Nd isotopic data for the whole rocks, to investigate the interaction of the magma with the marble wall rocks, the role of contamination in triggering sulfide saturation in the Jinchuan magma, and the formation of the ore deposit.

2. Geologic background

The Jinchuan ore-bearing mafic–ultramafic intrusion occurs in the eastern part of the Longshoushan terrane at the southern rim of the Alxa block, the westernmost part of the North China Craton (Fig. 1a). To the south the Longshoushan terrane is bounded by the North Qilian Orogenic Belt which formed by the collision between the Alxa block and the North Qilian block at ~ 450 Ma (Song et al., 2013). Regional metamorphism of green-schist facies related to the collision event has affected the Longshoushan terrane including the Jinchuan intrusion. The Jinchuan intrusion was emplaced as a sill-like body into a Paleoproterozoic suite of gneisses, migmatites, marbles and amphibolites, below an active intracratonic rift (Porter, 2016). Available data show that the 1.9–2.7 Ga amphibolites and granitic gneisses are the oldest rocks in this re-

gion (Geng et al., 2006; Gong et al., 2012). Abundant Paleozoic granites (Zeng et al., 2016) that are temporarily associated with the collision event between the Qilian Block and the North China Craton at ~ 450 Ma (Song et al., 2006) occur >50 km to the south (Fig. 1b). Li et al. (2005) reported a U–Pb SHRIMP zircon age of 827 ± 8 Ma for the Jinchuan intrusion. Zhang et al. (2010) reported a precise ID-TIMS zircon U–Pb age of 831.8 ± 0.6 Ma obtained on thermally annealed and chemically etched zircon grains from the intrusion.

The current surface exposure of the Jinchuan intrusion is ~ 6500 m long and ~ 500 m wide, with vertical downward extension >1100 m in its central part. The overall strike of the intrusion is northwest. The Jinchuan intrusion is divided into four segments by a series of northeast-trending strike-slip faults. They were numbered as III, I, II, and IV from west to east by the mine geologists based on the order of mine development (Fig. 1c). The Jinchuan intrusion is composed almost entirely of peridotites, including dunite, lherzolite, olivine websterite, and minor plagioclase lherzolite. The contacts between different types of rocks are transitional. Rare gabbros occur in the margins of the intrusion in a few places. Marble xenoliths are common along the northern margin and occur sporadically throughout the intrusion. They range up to 60 cm wide and at least 4 m long and have sharp contacts with the enclosing ultramafic rock. Their shapes are irregular, ranging from angular to round. In some places the ultramafic host adjacent to the xenoliths is only slightly altered; elsewhere it is largely to completely replaced by actinolite and chlorite. Mineralization occurs predominantly as disseminated to net-textured sulfides within the intrusion.

3. Samples and analytical methods

3.1. Samples

Lherzolite is the most abundant rock type in the Jinchuan intrusion. Thirteen lherzolite and five country rock samples used in this study were collected from drill cores and outcrops in the segments I and II of the Jinchuan intrusion where the most important Ni–Cu sulfide mineralization occurs.

Lherzolite contains 35 to 60% olivine and 10 to 50% pyroxenes (orthopyroxene $>$ clinopyroxene), plus minor Cr-spinel, hornblende, biotite and sulfide. Both orthopyroxene and clinopyroxene commonly occur as large oikocrysts enclosing small olivine crystals to form a poikilitic texture (Fig. 2a, b). The sulfide assemblages occur in the interstitial spaces of olivine or pyroxene cumulates (Fig. 2c). Both silicate and sulfide assemblages in the Jinchuan intrusion have experienced varying degrees of hydrothermal alteration and metamorphism. Olivine may be partially converted to serpentine + magnetite (Fig. 2b), and pyroxene may be altered to chlorite + tremolite + actinolite. Cr-spinel and disseminated sulfides are partially replaced by secondary magnetite in some samples.

The country rock marbles vary from white to pale green, and the mineral assemblages range from nearly pure dolomite (Fig. 2d), plus a very small amount of silicate minerals (mainly olivine) to (serpentinized) olivine–calcite $\pm$ clinozoisite $\pm$ tremolite. Grain size ranges from 0.1 to 2 mm.

3.2. Analytical methods

Whole-rock major and trace element compositions were determined at Chang'an University, Xi'an, the key laboratory of Crustal Dynamics, China Earthquake Administration, and the State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences, Guiyang, China. Major element abundances in whole-rocks were determined on fused glass disks using

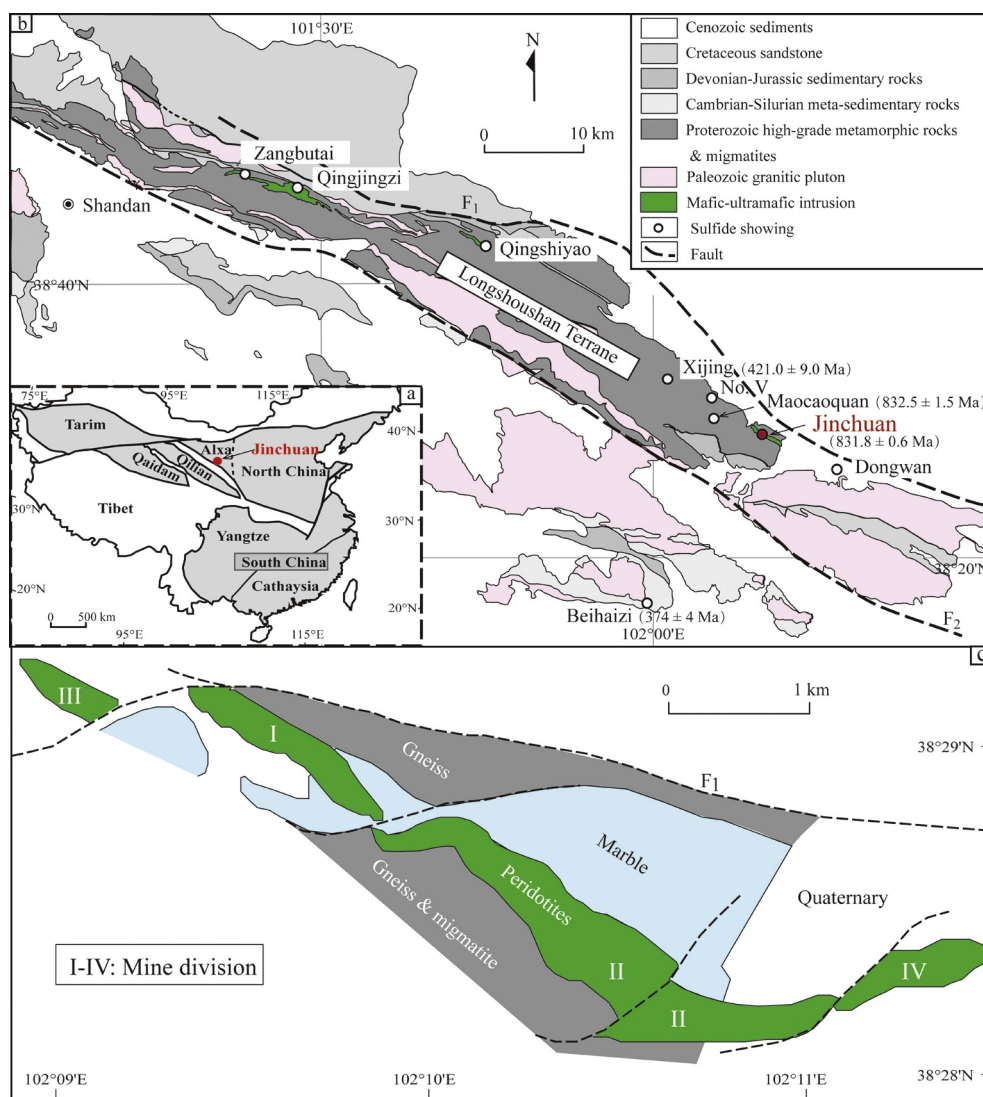


Fig. 1. The location of the Jinchuan Ni-Cu deposit in China (a), and simplified geologic maps of the Longshoushan terrane (b) and Jinchuan mafic-ultramafic intrusion (c), after Li and Ripley (2011). The zircon U-Pb ages for the mafic-ultramafic intrusive rocks are from Zhang et al. (2010), Jiao et al. (2012, 2017) and Duan et al. (2015).

X-ray fluorescence (XRF). Trace element compositions were measured by inductively coupled plasma mass spectrometry (ICP-MS). Rock powders were fused to form glass disks for XRF analysis, and acid-digested in steel-jacketed Teflon “bombs” to form solutions for ICP-MS analysis. The analytical uncertainties are between 5% and 15% relative, depending on the concentrations of elements in the samples.

Mg isotope analysis was performed at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences (Beijing) following the procedures similar to those established by Li et al. (2010), Teng et al. (2010) and Ke et al. (2016). Sample powders were precisely weighted into and dissolved in Savillex screw-up beakers with 3:1 mixture of HF and HNO₃ for 24 h at 160 °C. They were heated to dryness at 70–80 °C. The materials were re-dissolved with a 3:1 mixture of HCl and HNO₃ and were evaporated to dryness at 80 °C. Finally, the solutions were re-fluxed within the HNO₃ to achieve complete dissolution. Chemical separation of Mg was achieved by cation exchange chromatography with pre-cleaned resin (Bio-Rad 200–400 mesh AG50W-X8) twice. The solutions separated from the columns were heated to dryness and then dissolved in 3% HNO₃ for mass spectrometry. Magnesium isotope ratios were measured by the sample-standard bracketing method on a Neptune Plasma MC-ICP-MS with a ‘wet’

sample introduction system. Magnesium isotopes were analyzed in a low-resolution mode, with ²⁶Mg, ²⁵Mg, and ²⁴Mg measured simultaneously in separate Faraday cups (H3, IC, and L3). The sample-standard sequence was generally repeated 4–6 times to obtain better reproducibility and accuracy (Ke et al., 2016). The results are reported in the conventional δ notation that is defined as $\delta^X\text{Mg} = [({}^X\text{Mg}/{}^{24}\text{Mg})_{\text{sample}}/({}^X\text{Mg}/{}^{24}\text{Mg})_{\text{DSM3}} - 1] \times 1000$, where $X = 25$ or 26 , and the Dead Sea metal Mg standard (DSM3) is an international reference of solution made from pure Mg metal. The total procedural blanks were negligible and less than 10 ng, which is comparable with that of Teng et al. (2010). The internal precision of this study is $\pm 0.06\text{‰}$ (2SD) whereas the long-term external precision is $\pm 0.06\text{‰}$ (2SD) for $\delta^{26}\text{Mg}$ reported by Ke et al. (2016).

Whole-rock Sr and Nd isotopes were determined using a Triton thermal ionization mass spectrometer in the State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences, Guiyang, China. Mass fractionation corrections for Sr and Nd isotopic ratios were based on ${}^{86}\text{Sr}/{}^{88}\text{Sr} = 0.1194$ and ${}^{146}\text{Nd}/{}^{144}\text{Nd} = 0.7219$. The average isotopic ratios of the international standard (BCR-2) analyzed together with our samples are ${}^{87}\text{Sr}/{}^{86}\text{Sr} = 0.705178$ (the recommended value is 0.705) and ${}^{143}\text{Nd}/{}^{144}\text{Nd} = 0.512622$ (the recommended value is 0.512636).

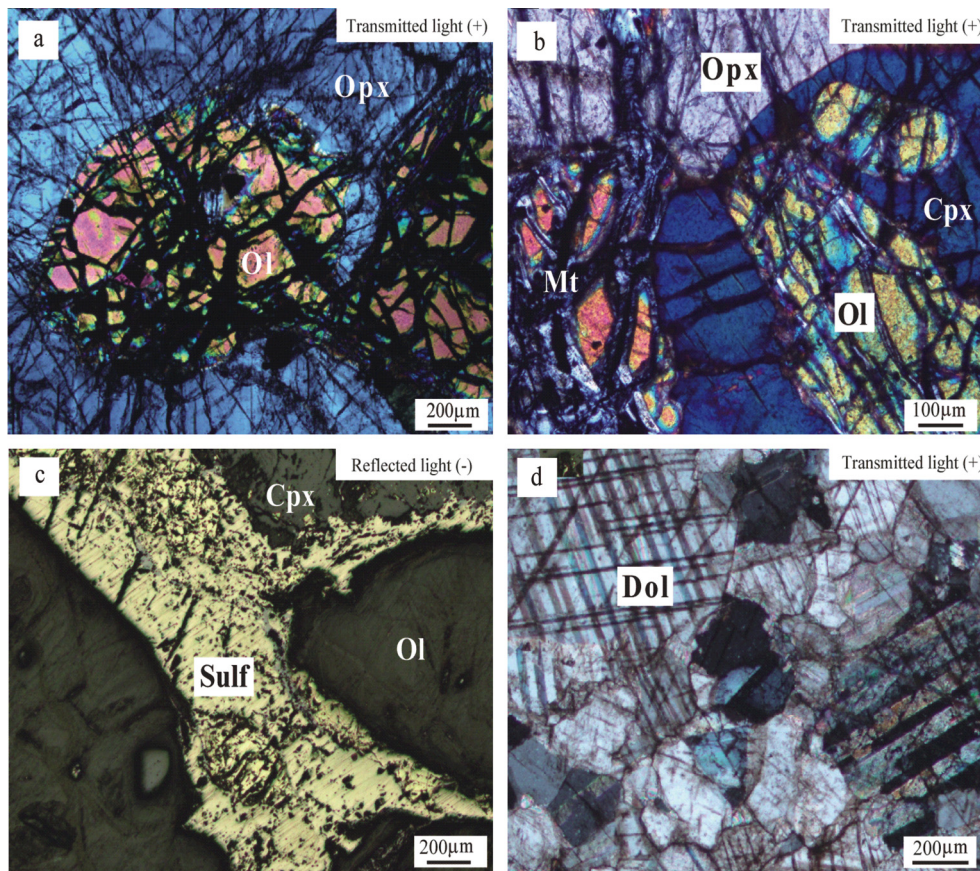


Fig. 2. Photomicrographs of lherzolites from the Jinchuan intrusion and marble country rocks. (a), (b) and (c) lherzolite, (d) marble. Abbreviations: Cpx = clinopyroxene, Dol = dolomite, Mt = magnetite, Ol = olivine, Opx = orthopyroxene, Sulf = sulfide.

4. Results

4.1. Major and trace elements

The whole-rock major and trace element compositions of the lherzolites from the Jinchuan Ni–Cu–(PGE) deposit and the marble country rocks are listed in Table S1 (Supplementary material). In order to compare the compositions of constituent minerals directly, we have normalized the original data to 100% anhydrous silicate compositions by correcting for loss-on-ignition for the lherzolites. As shown in Fig. 3, the compositions of lherzolites in the Jinchuan intrusion are mainly controlled by olivine, clinopyroxene and orthopyroxene plus minor amounts of hornblende and biotite. In the $\text{FeO}^{\text{Total}}$ versus MgO diagram, the rocks all plot above the trajectories of olivine and hornblende (Fig. 3c), consistent with the presence of magnetite and sulfide (Fig. 2b, c).

The marble samples JL325-360 and JL325-406 contain ~33.2 wt.% CaO, 19.6 wt.% MgO and 45.3 wt.% loss-on-ignition on average, values very close to those of pure dolomite. Contents of SiO_2 (~1.2 wt.%) and Al_2O_3 (~0.04 wt.%) are generally very low, indicating minimal amounts of silicate minerals. The other marbles contain relatively low CaO (24.0–29.8 wt.%) and loss-on-ignition (22.1–34.0 wt.%), as well as high MgO (18.7–23.5 wt.%) and SiO_2 (14.4–35.6 wt.%).

The chondrite-normalized rare earth elements (REE) and primitive mantle-normalized immobile trace element patterns of samples from the Jinchuan intrusion and marble country rocks are illustrated in Fig. 4. The lherzolites show moderate light REE enrichments relative to heavy REEs (Fig. 4a) and negative Nb anomalies (Fig. 4b). The marbles show strong light REE enrichments, positive U and Sr anomalies, and pronounced negative Nb anomalies

(Fig. 4). Except for U and Sr, the other trace element abundances in the marbles are 1–2 orders of magnitude lower than those in the lherzolites. Nb/U ratio vary from 11.57 to 24.04, which is significantly lower than that of E-MORB (~46, Sun and McDonough, 1989) and slightly lower than that of the lower crust (~25, Rudnick and Gao, 2014), but is significantly higher than that of the marbles from the country rocks (~0.76).

4.2. Mg isotopes

The Mg isotopic compositions for the olivine and pyroxene separates, and whole rocks of the lherzolites with marbles in this study are listed in Table 1. In the plot of $\delta^{25}\text{Mg}$ vs. $\delta^{26}\text{Mg}$ (Fig. 5), our samples fall within the predicted equilibrium mass-dependent fractionation line (0.521, Young and Galy, 2004). The $\Delta^{25}\text{Mg}'$ ($\Delta^{25}\text{Mg}' = \delta^{25}\text{Mg}' - 0.521\delta^{26}\text{Mg}'$, where $\delta^X\text{Mg}' = 1000 \ln(\delta^X\text{Mg}/1000 + 1)$, and X refers to mass 25 or 26, Young and Galy, 2004) values range between –0.008 and 0.015, very close to zero. These results indicate that Mg isotopes obey the mass-dependent fractionation law.

The $\delta^{26}\text{Mg}$ values for the olivine and pyroxene separates and for the whole rocks of lherzolites from the Jinchuan deposit range from –0.40‰ to –0.26‰, with a weighted mean value of $-0.31 \pm 0.06\text{‰}$ (2SD). This average value is slightly lighter than that of the global MORBs ($-0.25 \pm 0.06\text{‰}$, Teng et al., 2010) and the upper mantle ($-0.25 \pm 0.07\text{‰}$, Teng et al., 2010). Olivine separates define a range of $\delta^{26}\text{Mg}_{\text{olivine}}$ from –0.34‰ to –0.27‰, with an average value of $-0.30 \pm 0.06\text{‰}$ (2SD), which are almost identical to pyroxene in their Mg isotopic compositions ($\delta^{26}\text{Mg}_{\text{pyroxene}} = -0.40\text{‰}$ to -0.26‰) with an average value of $-0.32 \pm 0.06\text{‰}$ (2SD). The two whole rock lherzolite samples have a slightly heav-

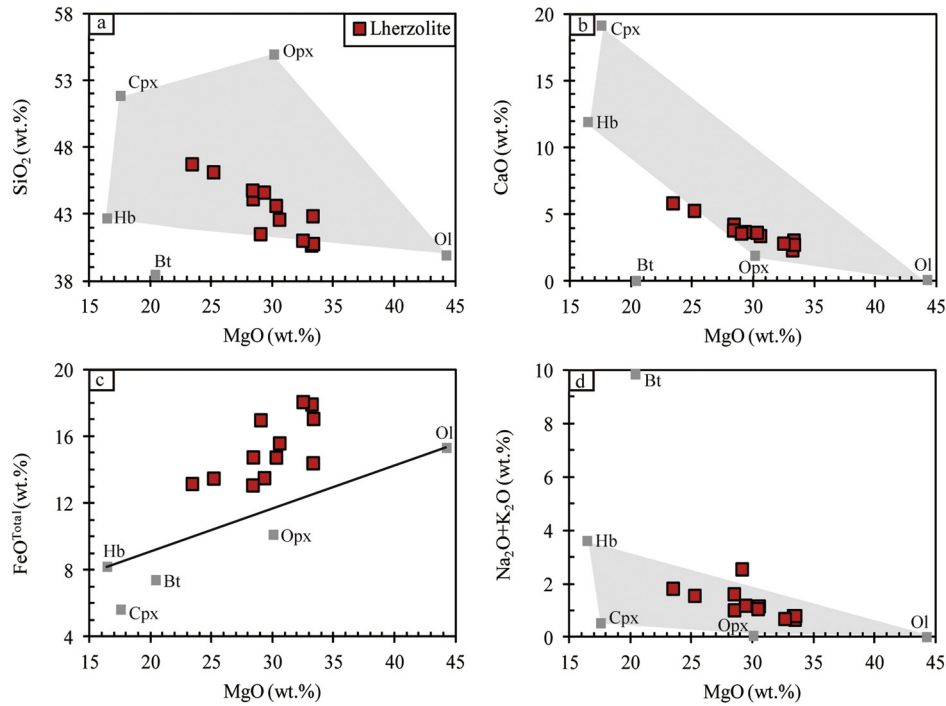


Fig. 3. Variations in major element compositions of rocks from the Jinchuan intrusion. The chemical compositions of olivine (Ol), clinopyroxene (Cpx), orthopyroxene (Opx), hornblende (Hb) and biotite (Bt) are from Chai and Naldrett (1992, 1994) and Li et al. (2004).

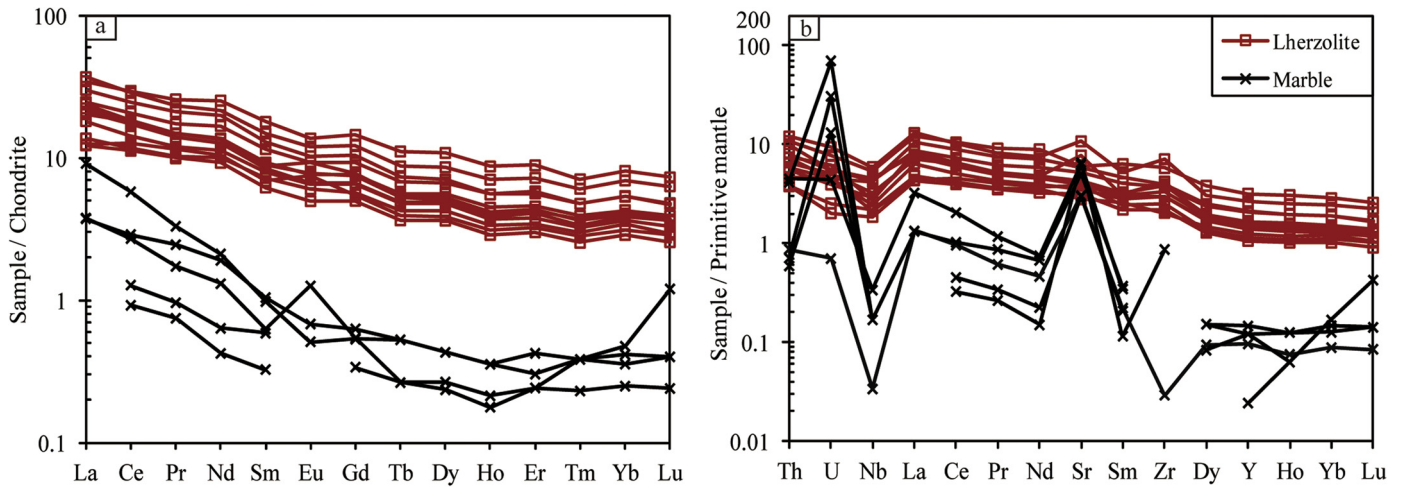


Fig. 4. Chondrite-normalized REE patterns (a) and primitive mantle-normalized trace element patterns (b) of the Jinchuan intrusion with marble country rocks. The chondrite and primitive mantle values for normalization are from Palme and O'Neill (2014).

ier Mg isotopic composition ($\delta^{26}\text{Mg} = -0.29 \pm 0.04\text{‰}$). Among these samples, the $\delta^{26}\text{Mg}$ values of one olivine and three pyroxene samples are out of the mantle range when error is considered (Fig. 6). These samples plot on the lower side of accepted mantle values represent anomalously low $\delta^{26}\text{Mg}$ values. The $\delta^{26}\text{Mg}$ values for the marbles from the country rocks of the Jinchuan deposit vary from -2.00‰ to -1.17‰ , with a weighted mean value of $-1.53 \pm 0.02\text{‰}$ (2SD).

4.3. Rb–Sr and Sm–Nd isotopes

The concentrations of Rb, Sr, Sm and Nd and their isotope ratios in the samples from the Jinchuan intrusion and the country rocks are given in Table 2. The calculated $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios ($t = 831.8$ Ma) and the ϵ_{Nd} values of the Jinchuan lherzolites range from 0.706615 to 0.714612 and from -8.29 to -6.13 , respectively, which are similar to previous results (Zhang et al., 2004). However,

the marble country rocks have relatively high $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios (0.713662 to 0.740069) and low ϵ_{Nd} values (-25.43 to -12.41).

5. Discussion

Serpentinization does not fractionate Mg isotopes (Dauphas et al., 2010; Beinlich et al., 2014). The $\delta^{26}\text{Mg}$ values of the olivine, pyroxene and lherzolites are invariant with LOI (Fig. 6a). The amount of LOI mainly reflects the degree of serpentinization, because these rocks are olivine cumulates and serpentine is the most important hydrous mineral replacing a large proportion of olivine in the samples. Previous analyses on lherzolite–harzburgite–dunite series peridotites worldwide yielded largely consistent $\delta^{26}\text{Mg}$ values, and hence suggest a homogeneous mantle ($\delta^{26}\text{Mg} = -0.25 \pm 0.07\text{‰}$, Handler et al., 2009; Bourdon et al., 2010; Teng et al., 2010). The light Mg isotopic ratios in the lherzolites from the Jinchuan deposit may result from magmatic differentiation, kinetic isotopic

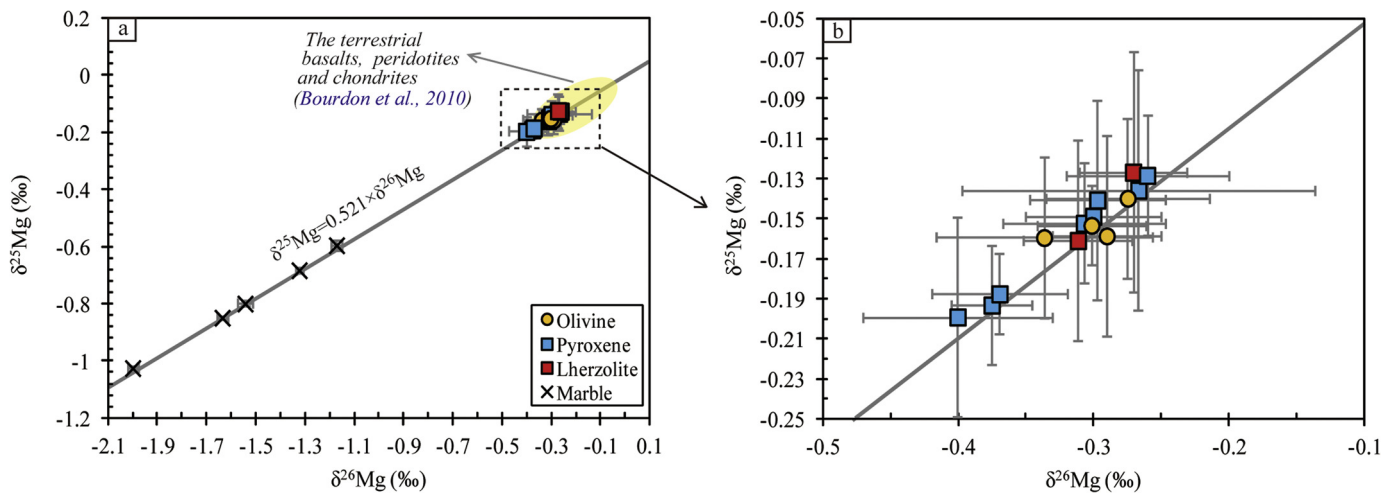


Fig. 5. Plot of $\delta^{26}\text{Mg}$ versus $\delta^{25}\text{Mg}$ for all samples reported in this study. The solid line represents the terrestrial fractionation line with a slope equal to 0.521 (Young and Galy, 2004). The yellow area represents the compositional range of terrestrial basalts, peridotites and chondrites as compiled by Bourdon et al. (2010). Error bars represent 2SD uncertainties.

Table 1

Mg isotopic compositions of mineral separates and whole rocks of lherzolites from the Jinchuan intrusion and marble country rocks.

Sample	Rock type	Mineral	$\delta^{25}\text{Mg}$ (‰)	2SD	$\delta^{26}\text{Mg}$ (‰)	2SD	$\Delta^{25}\text{Mg}'$ (‰)
ZK66-3	Lherzolite	Ol	−0.16	0.05	−0.29	0.04	−0.008
ZK73-9	Lherzolite	Ol	−0.16	0.04	−0.34	0.08	0.015
ZK49-3	Lherzolite	Ol	−0.14	0.04	−0.27	0.06	0.003
Y22U002	Lherzolite	Ol	−0.15	0.02	−0.30	0.04	0.003
ZK71-1	Lherzolite	Pyx	−0.15	0.03	−0.31	0.06	0.007
ZK73-4	Lherzolite	Pyx	−0.14	0.06	−0.27	0.13	0.003
ZK73-9	Lherzolite	Pyx	−0.14	0.05	−0.30	0.05	0.014
ZK73-11	Lherzolite	Pyx	−0.20	0.05	−0.40	0.07	0.009
ZK80-4	Lherzolite	Pyx	−0.15	0.01	−0.30	0.05	0.007
ZK87-1	Lherzolite	Pyx	−0.19	0.03	−0.37	0.03	0.002
Y22U001	Lherzolite	Pyx	−0.13	0.03	−0.26	0.06	0.007
Y22U06	Lherzolite	Pyx	−0.19	0.02	−0.37	0.05	0.005
JE1227-45	Lherzolite		−0.13	0.06	−0.27	0.04	0.014
JED93-146	Lherzolite		−0.16	0.05	−0.31	0.04	0.001
JL325-348	Marble		−0.60	0.02	−1.17	0.02	0.012
JL325-360	Marble		−0.85	0.01	−1.63	0.02	0.001
JL325-368	Marble		−1.03	0.01	−2.00	0.02	0.014
JL325-406	Marble		−0.80	0.01	−1.54	0.03	0.002
JL325-421	Marble		−0.69	0.01	−1.32	0.02	0.004

Notes: Ol – olivine; Pyx – pyroxene.

2SD = 2 times the standard deviation of the population of >20 repeat measurements of the standard during a session.

$\Delta^{25}\text{Mg}' = \delta^{25}\text{Mg}' - 0.521 \times \delta^{26}\text{Mg}'$, where $\delta^X\text{Mg}' = 1000 \times \ln(\delta^X\text{Mg}/1000 + 1)$ (Young and Galy, 2004).

fractionation within the magma chamber and/or around the contact boundaries (Teng et al., 2011), or assimilation of wall rocks during ascent. In this section, we first explore the possible mechanisms responsible for the light Mg isotopic signatures. Then we discuss the magma-marble interaction during the formation of the Jinchuan deposit based on the observations from Mg–Sr–Nd isotope geochemistry of the lherzolites and marble country rocks.

5.1. The origin of light magnesium isotopic compositions in the Jinchuan intrusion

Olivine and pyroxene separates from the lherzolites in the Jinchuan intrusion have nearly identical average $\delta^{26}\text{Mg}$ values. The $\delta^{26}\text{Mg}$ values are invariant with MgO , SiO_2 and CaO (Fig. 6b–d), indicating that the variable accumulation of olivine, clinopyroxene and orthopyroxene after fractional crystallization in the Jinchuan deposit has negligible effects on the Mg isotope fractionation at bulk-rock scale.

Olivine and pyroxene in sample ZK73-9 have narrow inter-mineral Mg isotope fractionation, with $\Delta^{26}\text{Mg}_{\text{pyx-ol}} = 0.04\text{‰}$, indicating that Mg isotopes reached equilibrium in the lherzolites. It

rules out chemical diffusion driven kinetic fractionation as a cause of the isotopically light lherzolites. If Mg diffused from basaltic magma to the marble country rocks, Mg isotopic composition of the basaltic magma should be shifted towards light values. This is the opposite to the lherzolites from the Jinchuan intrusion. Therefore, both chemical and thermal diffusion appear unable to create the light Mg isotopic compositions in the Jinchuan intrusion.

Crustal contamination by silicate wall rocks are unlikely to produce such a light Mg isotopic compositions in the Jinchuan intrusion because crustal materials generally have low-Mg content compared to the highly magnesian melt. The bulk continental crust has an average $\delta^{26}\text{Mg} = -0.24 \pm 0.07\text{‰}$, which is indistinguishable from the average mantle ($\delta^{26}\text{Mg} = -0.25 \pm 0.07\text{‰}$) (Teng et al., 2010). The average $\delta^{26}\text{Mg}$ values of the upper crust ($-0.22 \pm 0.10\text{‰}$, Li et al., 2010), middle crust ($-0.21 \pm 0.07\text{‰}$), and lower crust ($-0.26 \pm 0.06\text{‰}$, Yang et al., 2016) are slightly heavier than those of olivine, pyroxene and whole rock samples in the Jinchuan deposit. In this case, incorporation of continental crustal components would make Mg isotopes heavier than the mantle reservoir,

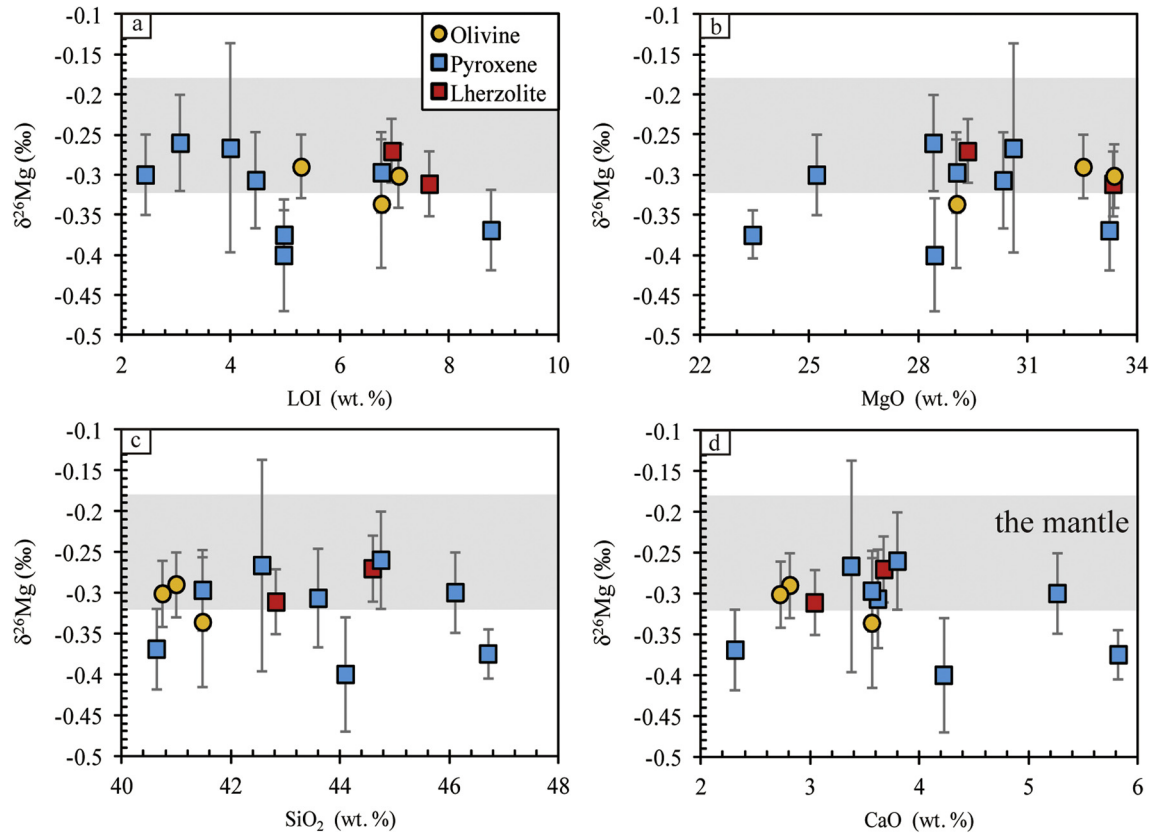


Fig. 6. The plots of $\delta^{26}\text{Mg}$ versus LOI, MgO, SiO_2 and CaO for the Jinchuan intrusion. The gray shaded area represents the Mg isotopic compositional range of the average mantle ($\delta^{26}\text{Mg} = -0.25 \pm 0.07\text{‰}$, Teng et al., 2010).

which is in striking contrast to the overall light Mg isotopic signatures in the Jinchuan deposit.

Hence, the shift in $\delta^{26}\text{Mg}$ of the lherzolites to lower values could be a reflection of contamination of Mg-rich carbonate wall rocks during magmatic ascent. The wall rocks, marbles and calc-silicates, have lighter Mg isotopic compositions (-2.00‰ to -1.17‰), and marble xenoliths have been found throughout the intrusion; for these reasons the light $\delta^{26}\text{Mg}$ values in the separated minerals and whole rocks of lherzolites from the Jinchuan intrusion may be caused by carbonate wall rock assimilation during magma ascent, differentiation, and emplacement.

5.2. Source mantle characteristics and crustal contamination

Negative Nb anomalies relative to U and La (Fig. 4b), and $\text{Nb/U} = 11.57\text{--}24.04$ in the Jinchuan intrusion indicate that its parental magma was contaminated by crustal materials. The low abundance of Th and U compared with the light REE (Table S1) suggest that rocks from the lower continental crust were assimilated (Lehmann et al., 2007). The zircon Hf isotope ($\epsilon_{\text{Hf}} = -4$ to -11 , Tang et al., 2014; Duan et al., 2016), Os isotope ($\gamma_{\text{Os}} = 30\text{--}320$, Zhang et al., 2004; Yang et al., 2005, 2008) and sulfur isotope ($\delta^{34}\text{S} = -2\text{‰}\text{--}8\text{‰}$, $\Delta^{33}\text{S} = 0.12\text{‰}\text{--}2.67\text{‰}$, Ripley et al., 2005; Duan et al., 2016) also indicated crustal contamination was involved in the generation of the Jinchuan deposit. Based on mineralogy, major and trace elements, and C–O isotopes, Lehmann et al. (2007) proposed that the Jinchuan parental magma assimilated granitic rocks in the lower crust, and when the magma reached its present level in the middle to upper crust, the magma assimilated marbles.

Our data have large variation in $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios and relatively high ϵ_{Nd} values compared to Duan et al. (2016)'s data, but they are similar to the data of Zhang et al. (2004) (Fig. 7a). The difference

of Nd isotope between our data and that of Duan et al. (2016) is difficult to evaluate. It may be caused by the instrumental bias, and different sample location or depth. We used a Triton Thermal Ionization Mass Spectrometer, whereas Duan et al. (2016) used a MC-ICP-MS instrument. The vertical and horizontal distances between our samples and the samples of Duan et al. (2016) are from 50 to 2200 m. Different parts of the Jinchuan intrusion may have formed from different pulses of magma that experienced different degrees of contamination with the lower crust, and partially account for the observed variation of ϵ_{Nd} values.

To identify and constrain the contamination involved in the formation of the Jinchuan deposit, we modeled Mg, Sr, and Nd isotopic variations. In our calculations we used the average MgO content in the Jinchuan parental magma estimated by Chai and Naldrett (1992) and Li and Ripley (2011), and the average MgO content and Mg–Sr–Nd isotopic compositions of the marbles. The source mantle Sr–Nd isotope composition is inferred from the interception of the Jinchuan Sr–Nd isotope array with the mantle array. As shown in Fig. 7a, the Sr–Nd isotope compositions of the Jinchuan intrusion indicate 10 to 20 wt.% bulk assimilation of the lower continental crust by a mantle-derived magma with ϵ_{Nd} close to -2 and $(^{87}\text{Sr}/^{86}\text{Sr})_i$ close to 0.705. Although Sr isotopic compositions are more susceptible to the post-magmatic hydrothermal alteration than Nd isotopes, whole-rock Sr isotopes are more sensitive to marble contamination due to the high Sr concentration and low Nd concentration in the marbles. The large variation in $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios of the Jinchuan intrusion may be due to the large amount of marble contamination. Large Mg isotope fractionation has been recognized during weathering of silicate rocks and carbonates, but the Mg isotope values of the weathered rocks tend to be heavier instead of lighter (Teng et al., 2010). The Mg–Nd–Sr isotopic data collected in our study indicate that the addition of $<10\%$ of marble to the Jinchuan magma is inferred to have

Table 2
Whole-rock Sr–Nd isotopic compositions in the Jinchuan intrusion and marble country rocks.

Sample	Rock type	Rb	Sr	Sm	Nd	$^{87}\text{Rb}/^{86}\text{Sr}$	$^{87}\text{Sr}/^{86}\text{Sr}$	2σ	$^{147}\text{Sm}/^{144}\text{Nd}$	$^{143}\text{Nd}/^{144}\text{Nd}$	2σ	$(^{87}\text{Sr}/^{86}\text{Sr})_i$	$\epsilon_{\text{Nd}}(t)$
ZK71-1	Lherzolite	7.484	94.04	1.42	6.458	0.230341	0.711596	5E–06.000000	0.132911	0.511974	8E–06.000000	0.708859	–6.17
ZK73-9	Lherzolite	6.016	87.32	1.288	6.016	0.199358	0.708983	7E–06.000000	0.129413	0.511958	9E–06.000000	0.706615	–6.13
ZK73-11	Lherzolite	8.766	115.82	1.764	7.914	0.219013	0.709249	7E–06.000000	0.134733	0.511968	5E–06.000000	0.706647	–6.49
ZK87-1	Lherzolite	18.766	235.8	2.272	10.17	0.230402	0.714141	5E–06.000000	0.135037	0.511929	4E–06.000000	0.711404	–7.28
Y22U006	Lherzolite	10.386	76.98	1.208	5.322	0.337663	0.718624	9E–06.000000	0.127894	0.511926	1.1E–050	0.714612	–6.59
JE1227-45	Lherzolite	14.46	129.8	1.998	9.402	0.322409	0.710730	5E–06.000000	0.128451	0.511882	7E–06.000000	0.706900	–7.49
JE1227-146	Lherzolite	6.11	61.16	1.114	4.764	0.289171	0.712298	4E–06.000000	0.141344	0.511912	7E–06.000000	0.708862	–8.29
JL325-348	Marble	0.3	137.5	0.15	1	0.006333	0.740145	8E–06.000000	0.090654	0.511244	7E–06.000000	0.740069	–15.94
JL325-360	Marble	1	144	0.05	0.2	0.020142	0.732993	1.4E–050	0.151090	0.511231	1E–05.000000	0.732753	–22.64
JL325-368	Marble	0.7	110.5	0.16	0.9	0.018370	0.731111	8E–06.000000	0.107449	0.511517	1E–05.000000	0.730892	–12.41
JL325-406	Marble	0.526	67.14	0.096	0.618	0.022681	0.713931	1E–05.000000	0.093881	0.511232	8E–06.000000	0.713662	–16.53
JL325-421	Marble	0.1	65.3	0.09	0.3	0.004435	0.716671	1.4E–050	0.181309	0.511253	9E–06.000000	0.716618	–25.43

Notes: ϵ_{Nd} is the derivation in parts per 10,000 of the initial ratio from that of a chondritic reservoir at the crystallization age of 831.8 Ma. Estimation at $t = 831.8$ Ma was calculated with $\lambda = 1.42 \times 10^{-11}$.

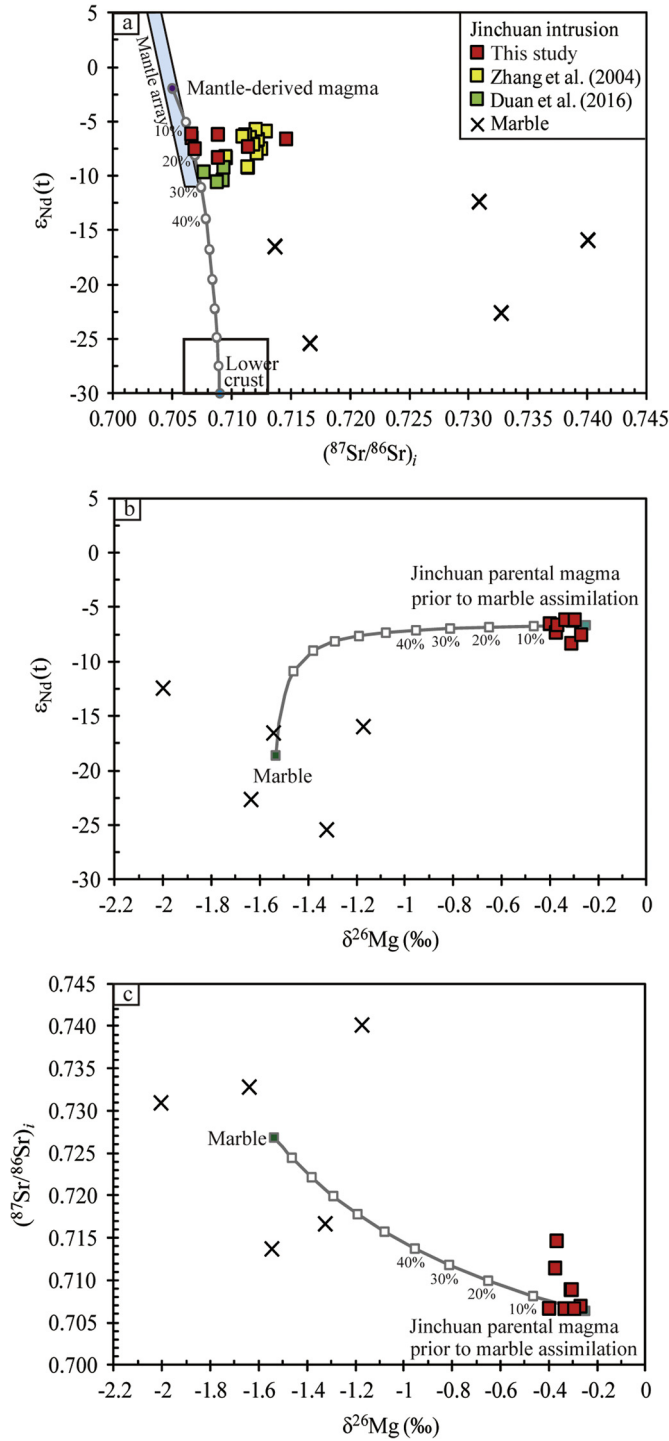


Fig. 7. The plots of ϵ_{Nd} versus $(^{87}\text{Sr}/^{86}\text{Sr})_i$ (a), ϵ_{Nd} versus $\delta^{26}\text{Mg}$ (‰) (b), and $(^{87}\text{Sr}/^{86}\text{Sr})_i$ versus $\delta^{26}\text{Mg}$ (‰) (c). The mantle Sr–Nd isotope array is from Zindler and Hart (1986). The Sr–Nd isotope compositions of the lower crust are from Jahn et al. (1999). The Sr and Nd element concentrations in the lower crust are from Rudnick and Gao (2014). The MgO content in the Jinchuan parental magma is from Chai and Naldrett (1992) and Li and Ripley (2011). The Mg isotope compositions of the average mantle and the lower crust are from Teng et al. (2010) and Yang et al. (2016), respectively. The average MgO content and Mg–Sr–Nd isotopic compositions of marbles are from this study.

occurred after the magma had experienced contamination with the bulk lower continental crust (Fig. 7b, c). The samples show no correlation between $(^{87}\text{Sr}/^{86}\text{Sr})_i$ and $\delta^{26}\text{Mg}$, and some samples plot away from the mixing line (Fig. 7c), which may have resulted from the different percentages of carbonate mineral phases in the marbles, such as calcite, dolomite and brucite, or from postmagmatic hydrothermal alteration, or both. The modeling results described above are for bulk assimilation. Selective assimilation with a higher brucite/dolomite ratio than that of the marble will reduce the amounts of assimilation required to produce the contaminated magma.

5.3. The formation mechanism of Jinchuan deposit

Olivine Fo and Ni contents indicate that fractional crystallization did not play a major role in triggering sulfide saturation in the Jinchuan magma (Li and Ripley, 2011; Duan et al., 2016). Minor elements and Mg–Sr–Nd isotopes reveal that crustal contamination involving siliceous and marble wall rocks may have triggered sulfide segregation in the Jinchuan intrusion. Experimental studies have shown that the addition of siliceous materials can cause a basaltic magma to become saturated in sulfide (see summary in Naldrett, 2011). Oxidation of magma via assimilation of carbonate country rocks can also lead to sulfide saturation (Lehmann et al., 2007; Ganino et al., 2014). At relatively reduced conditions, where sulfide is the predominant sulfur species in a magma, oxidation will lead to a decrease in FeO content which in turn will decrease the sulfur concentration needed to achieve sulfide liquid saturation (Ripley and Li, 2013).

The oxidation state of the Jinchuan intrusion varied between quartz-fayalite-magnetite (QFM) + 0.3 and QFM + 1, indicating that the magmas that produced the Jinchuan deposit were more oxidized (Duan et al., 2016) than the Voisey's Bay deposit in Labrador, Canada, which represents the most reduced magmatic sulfide deposit known worldwide (Barnes et al., 2013). The more oxidized nature of the Jinchuan intrusion is believed to have been caused by magma interaction with marble country rocks and CO_2 liberated by decarbonation reactions (Lehmann et al., 2007). The injection of a large amount of CO_2 into the magma system increases the CO_2/CO ratio, and this may have made the magma more oxidized (Ganino et al., 2008).

A possible scenario for the formation of the Jinchuan Ni–Cu–(PGE) sulfide ore deposit therefore is: (1) assimilation of lower crustal wall rocks, which decreased the sulfur solubility by means of input of a large amount of SiO_2 and Al_2O_3 (Song et al., 2012) and explains the multiple sulfur isotope compositions of Jinchuan sulfides (Duan et al., 2016); (2) assimilation of marble country rocks, which oxidized the magma by means of addition of CO_2 -rich fluids, decreased the sulfide solubility further by means of the reduced Fe^{2+} content of the magma, and facilitated the segregation of an immiscible sulfide liquid (Lehmann et al., 2007). Therefore, the assimilation of siliceous crustal materials and marbles may have been instrumental in the formation of the Jinchuan ore deposit.

5.4. Implication for the genesis of the magmatic deposits

Geochemical studies clearly indicate that crustal components are commonly involved in the genesis and evolution of magmatic Ni–Cu–PGE sulfide deposit (e.g. Naldrett, 2011; Ripley and Li, 2013). Magma–carbonate/marble interaction, in particular, is not a rare process and its occurrence has been suggested and demonstrated for several magmatic systems. Ganino et al. (2014) proposed that the Aguablanca Ni–Cu deposit in Europe formed by contamination involving terrigenous sediments and carbonates. Ganino et al. (2008) and Tang et al. (2017) propose that the

Panzhihua-type Fe–Ti–V oxide deposits in the Emeishan large igneous province formed due to carbonate assimilation by basaltic magma.

The results from this study show that marble assimilation by basaltic magma is an efficient way to produce sulfide saturation in magmas. The Jinchuan deposit remains as an isolated occurrence despite continuous regional exploration for many decades (Li and Ripley, 2011). Previous studies have shown that the Jinchuan intrusion is the product of continental rift-related basaltic magmatism which took place during the early stage of Rodinia breakup. The western parts of the North China Block have the best potential for new discoveries of Jinchuan-type Ni–Cu ore deposit (e.g., Li and Ripley, 2011; Tang et al., 2014). In addition to locating mafic-ultramafic intrusions with ages similar to that of the Jinchuan intrusion, such as the Maocaoquan intrusion (832.5 ± 1.5 Ma, Jiao et al., 2012), we suggest that country rocks with similar lithogeochemical characteristics to those of the Jinchuan intrusion are also important for successful exploration.

Mg isotopes, in combination with other isotopic systems, can be used to constrain mantle sources due to limited Mg isotope fractionation during silicate magma differentiation. Different ratios of Mg isotopes in mantle rocks, clastic sedimentary rocks and carbonates make Mg isotopes an effective tool in the evaluation of crustal contamination. Mg isotopes may supplement traditional methods such as radiogenic isotope and trace element data, particularly in cases in which a high fraction of crustal material has been added (Brewer et al., 2018). Studies of Mg isotopes have provided new information on ore genesis in mafic magma systems. Magnesium isotopes are a reliable tracer of magma–carbonate interaction and may potentially be used to trace ore-forming processes (i.e., source composition and crustal contamination) in mafic magmatic systems.

6. Conclusions

We have demonstrated that magnesium isotope variations are a reliable tracer of magma–carbonate interaction. This is mainly because Mg is a major element in dolomitic marbles and the isotopes of Mg in marbles are significantly different from those in mantle-derived magma. Furthermore, Mg isotopes may provide a tool for detecting crustal assimilation in settings where commonly-used radiogenic isotopes are ambiguous due to lack of isotopic contrast between the crust and primary magmas. The combination of integrated Mg–Sr–Nd isotopes can thus be used to constrain the origin, magma evolution, and the formation mechanism of many magmatic deposits. The Mg–Sr–Nd isotopic compositions of the Jinchuan intrusion are consistent with 10–20% bulk assimilation of the lower continental crust by a mantle-derived magma, followed by less than 10% bulk assimilation involving dolomitic marbles. These two processes may have triggered sulfide segregation which lead to the formation of the Jinchuan ore deposit. Marble assimilation by basaltic magma is an efficient mechanism for generating sulfide saturation in mafic magmas, which in turn could lead to the formation of Ni–Cu–PGE sulfide ore deposits. Locating intrusions with a similar age to that of the Jinchuan intrusion, and country rocks with similar lithogeochemical characteristics, are crucial in the exploration for additional Jinchuan-type deposits.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epsl.2018.09.008>.

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